RCS05 | Instructor Guide

CTS Cohort 6 | CTS Day 13 | 2026-07-27 | 10:00 AM-3:00 PM | 6 students

Advanced Electrical: Switch, Light, and Troubleshooting

Core rule: A professional does not guess. A professional checks, protects, and verifies.

Big idea: Troubleshooting is disciplined evidence gathering: verify the source state, follow the path, find the first failed checkpoint, correct one issue, and retest.

Architecture of the day

Time	Instructor move
10:00-10:15	Arrival, PPE, retrieval warm-up
10:15-10:30	Morning quiz
10:30-10:50	Quiz review and energy-boundary check
10:50-11:20	Switch-controlled lighting path model
11:20-12:00	Splice, pigtail, and termination demonstration
12:00-12:30	Troubleshooting ladder guided practice and lab release
12:30-1:00	Lunch and controlled reset
1:00-2:25	Switch/light install and inserted-fault lab
2:25-2:45	Correction, retest, evidence, cleanup
2:45-3:00	Exit ticket and Day 6 preview

Learning targets

- Trace a de-energized switch-controlled lighting path
- Install a training switch and light outlet box
- Use visual inspection and continuity to diagnose inserted faults
- Document the fault, evidence, correction, and retest

Materials and setup

- Six isolated switch/light training stations
- Switch boxes, light outlet boxes, training switches, light points, cable, and pigtail materials
- Approved line-switch-load-neutral diagrams
- Meters reserved for isolated continuity testing
- Instructor-only fault cards and reset controls
- PPE, labels, hand tools, and evidence devices

Instructor talking points

- Retrieve PV101 Day 6 hidden-fuse troubleshooting logic and apply it to residential training circuits.
- Trace line -> switch -> load -> neutral return and the separate grounding path.
- Model safe splice preparation, pigtails, box fill awareness, labeling, and device orientation at training level.
- Demonstrate the troubleshooting ladder with no energized student work.
- Connect all work to the capstone requirement: two outlets, one switch, and one light point.

Whiteboard plan

1. Expected path
2. Known source state
3. First failed checkpoint
4. One correction
5. Retest and document

Lab facilitation

Students add one training switch box and one light outlet box to the capstone wall, complete de-energized terminations, then diagnose an instructor-inserted open-circuit or misconnection using visual and continuity evidence.

Lab sequence

6. Energy-control check
7. Review diagram
8. Lay out boxes
9. Mount and route
10. Prepare splice/pigtails
11. Terminate switch and light path
12. Instructor inspection
13. Receive faulted state
14. Predict and test checkpoints
15. Correct one fault
16. Retest
17. Document and reset

Safety boundaries

- Training mock-up only. Keep every circuit de-energized, isolated from premises wiring, and under instructor control. This class does not authorize unlicensed field electrical work.
- Instructor verifies isolation before testing

- No student energized diagnostics
- Continuity only on isolated conductors
- Replace damaged devices or conductors
- Close and label the training system after approval

Evidence targets

- Lighting-path diagram
- Switch/light installation QC
- Troubleshooting log
- Before/after test results
- Instructor verification

Quiz answer key

1. What is an open circuit? A broken or incomplete current path.

2. What conductor path is controlled by a basic switch? The ungrounded or hot path to the load.

3. What should troubleshooting begin with? A known expected path and verified source/energy state.

4. What does continuity indicate? Whether an unbroken conductive path exists; it does not prove safe energized operation.

5. Why make one correction at a time? To preserve cause-and-effect evidence.

6. What is a pigtail? A short conductor used to connect a splice to a device or bonding point.

7. May students use continuity on an energized circuit? No.

8. What is the first failed checkpoint? The first point along the expected path where the observed result stops matching the expected result.

9. What capstone devices are required? Two outlets, one switch, and one light point.

10. What completes a professional repair? Correction, retest, documentation, and instructor approval.

Exit-ticket look-fors

- Draw line-switch-load-neutral
- Name the first failed checkpoint
- State the energy boundary
- Describe the evidence that proved your correction

Google Classroom post

RCS Day 5 - Advanced Electrical: Switch, Light, and Troubleshooting

Today we focused on troubleshooting is disciplined evidence gathering: verify the source state, follow the path, find the first failed checkpoint, correct one issue, and retest. Students completed guided instruction, hands-on capstone work, verification, evidence capture, and cleanup. Submit your worksheet, lab evidence, and exit ticket before leaving.

Source and safety references

- **Canonical RCS skeleton:** /CANONICAL/00_INBOX/Downloads/rcs_curriculum_plan.md
- **Cohort 6 RCS01 source:** /CANONICAL/02_PROJECTS/CTS_Cohort_06/03_DAILY/D09_RCS01_2026-07-20/Source/lesson.md
- **OSHA PPE:** https://www.osha.gov/personal-protective-equipment/construction
- **OSHA eye and face protection:** https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.102
- **OSHA circuit lockout/tagging:** https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.417
- **Oregon electrical codes:** https://www.oregon.gov/bcd/codes-stand/pages/electrical.aspx
- **Oregon permits:** https://www.oregon.gov/bcd/lbdd/pages/oregon-permits.aspx

RCS05 | Morning Quiz - Answer Key

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1. What is an open circuit?

A broken or incomplete current path.

2. What conductor path is controlled by a basic switch?

The ungrounded or hot path to the load.

3. What should troubleshooting begin with?

A known expected path and verified source/energy state.

4. What does continuity indicate?

Whether an unbroken conductive path exists; it does not prove safe energized operation.

5. Why make one correction at a time?

To preserve cause-and-effect evidence.

6. What is a pigtail?

A short conductor used to connect a splice to a device or bonding point.

7. May students use continuity on an energized circuit?

No.

8. What is the first failed checkpoint?

The first point along the expected path where the observed result stops matching the expected result.

9. What capstone devices are required?

Two outlets, one switch, and one light point.

10. What completes a professional repair?

Correction, retest, documentation, and instructor approval.

Scoring guidance

One point per item. Accept technically equivalent wording. Use errors to reteach before lab release. Safety-boundary errors must be corrected explicitly before the student begins the related task.

RCS05 | Student Packet

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Big idea: Troubleshooting is disciplined evidence gathering: verify the source state, follow the path, find the first failed checkpoint, correct one issue, and retest.

Learning targets

- Trace a de-energized switch-controlled lighting path
- Install a training switch and light outlet box
- Use visual inspection and continuity to diagnose inserted faults
- Document the fault, evidence, correction, and retest

Today's run

10:00-10:15 | Arrival, PPE, retrieval warm-up

10:15-10:30 | Morning quiz

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Key words

line | load | switch leg | open circuit | short circuit | fault | continuity | splice | pigtail | verification

Guided worksheet

1. Draw the switch-controlled light path.

_____

_____

2. Predict three faults that could create an open circuit.

_____

_____

3. Write the safe continuity-test sequence.

_____

_____

4. Record expected and observed results at each checkpoint.

_____

_____

5. Explain why changing two things at once weakens the diagnosis.

_____

_____

Afternoon lab mission

Students add one training switch box and one light outlet box to the capstone wall, complete de-energized terminations, then diagnose an instructor-inserted open-circuit or misconnection using visual and continuity evidence.

Safety boundary

- Training mock-up only. Keep every circuit de-energized, isolated from premises wiring, and under instructor control. This class does not authorize unlicensed field electrical work.
- Instructor verifies isolation before testing
- No student energized diagnostics
- Continuity only on isolated conductors
- Replace damaged devices or conductors
- Close and label the training system after approval

Build / test sequence

[ ] Energy-control check

[ ] Review diagram

[ ] Lay out boxes

[ ] Mount and route

[ ] Prepare splice/pigtails

[ ] Terminate switch and light path

[ ] Instructor inspection

[ ] Receive faulted state

[ ] Predict and test checkpoints

[ ] Correct one fault

[ ] Retest

[ ] Document and reset

Evidence record

[ ] Lighting-path diagram: _____

[ ] Switch/light installation QC: _____

[ ] Troubleshooting log: _____

[ ] Before/after test results: _____

[ ] Instructor verification: _____

Reflection and exit ticket

Draw line-switch-load-neutral

_____

_____

Name the first failed checkpoint

_____

_____

State the energy boundary

_____

_____

Describe the evidence that proved your correction

_____

_____

Instructor verification

Status: [ ] Approved [ ] Correct and recheck [ ] Needs follow-up

Instructor initials: _____ Date/time: _____

RCS05 | Morning Quiz

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Advanced Electrical: Switch, Light, and Troubleshooting

Name: _____ Date: _____

Answer from your own understanding. This is a teaching check, not a trap.

1. What is an open circuit?

_____

_____

2. What conductor path is controlled by a basic switch?

_____

_____

3. What should troubleshooting begin with?

_____

_____

4. What does continuity indicate?

_____

_____

5. Why make one correction at a time?

_____

_____

6. What is a pigtail?

_____

_____

7. May students use continuity on an energized circuit?

_____

_____

8. What is the first failed checkpoint?

_____

_____

9. What capstone devices are required?

_____

_____

10. What completes a professional repair?

_____

_____